

Notes On What I Learned In Making The Project

Logic Design

- Binary (0,1)

Bits are binary digits 4 bits = 1010

- Digital systems take inputs and outputs

- Our number system

	0-9	0-9	0-9	Binary 0-1	Binary 0-1	Binary 0-1
	8	2	5	0	1	0
X						
	10^3	10^2	10^1	2^2	2^1	2^0
	8	2	5	0	0	2

Hexadecimal is a way of understanding binary

0-9 10 11 12 13 14 15
A B C D E F

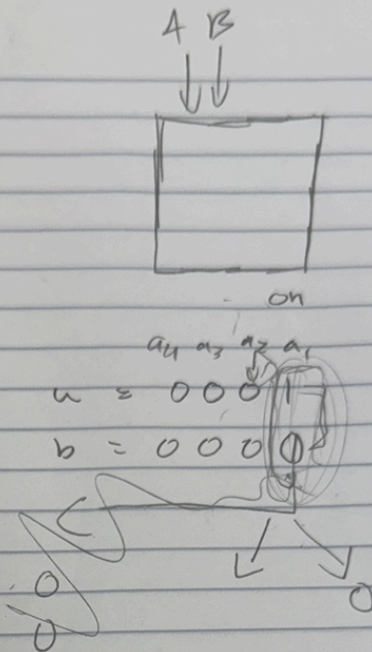
0	B	4
16^2	16^1	16^0
0	176	4
1	8	0

Binary addition

0110 6
0111 7

$$\begin{array}{r} 0110 \\ + 0111 \\ \hline 1101 \end{array}$$

$$\begin{array}{r}
 1101 \quad 13 \\
 -0010 \quad 2 \quad 11 \\
 \hline
 \end{array}$$



HV

- module full_adder

define a circuit called full_adder

defining inputs as logic

input logic A, A is an input in C1

;

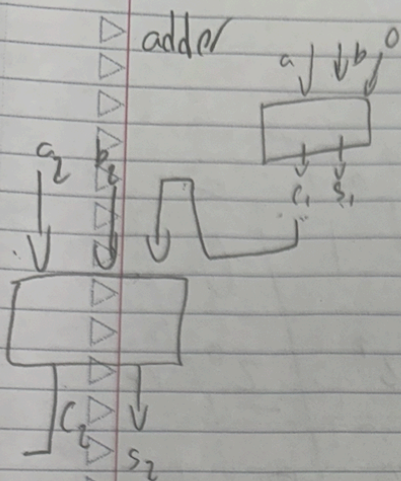
assign *Output* = *Inputs* ^ xor & and | or

taking that logic

module four_bit_adder

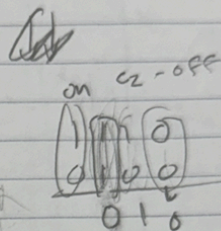
input logic [3:0] A, a basically makes A0-A3

Full_adder FA0 C

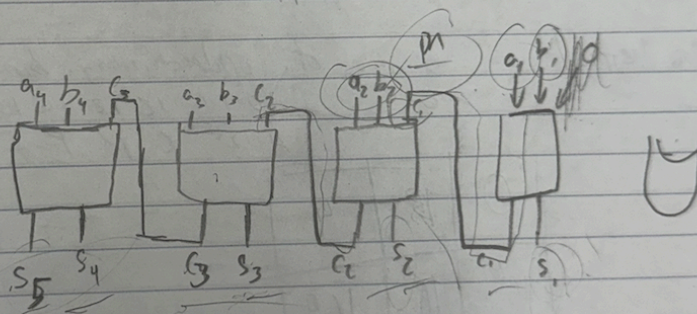


a	b	c ₁	s ₁
0	1	0	1
1	1	1	0
1	0	0	1

c is the carry to the next adder



4 bit adder logic



(2) (8) (8) (8)

111 ✓

110 ✓

101 ✓

011

001

$$s_1 = a_1 \oplus b_1$$

$$c_1 = a_1 \& b_1$$

$$s_2 = (a_2 \oplus b_2) \oplus c_1$$

$$c_2 = (a_2 \& b_2) \vee (b_2 \& c_1) \vee (a_2 \& c_1)$$

$$s_3 =$$